

4.10 Pin definition

Table 11. Legend/abbreviations used in the pinout table

Name	Abbreviation	Definition
Pin name	Unless otherwise specified in brackets below the pin name, the pin function during and after reset is the same as the actual pin name	
Pin type	S	Supply pin
	I	Input only pin
	I/O	Input / output pin
I/O structure	FT	5 V tolerant I/O
	TT	3.6 V tolerant I/O
	B	Dedicated BOOT0 pin
	NRST	Bidirectional reset pin with embedded weak pull-up resistor
	Option for TT or FT I/Os	
	_a ⁽¹⁾	I/O, with Analog switch function supplied by V _{DDA}
	_c	I/O, USB Type-C PD capable
	_d	I/O, USB Type-C PD Dead Battery function
	_f ⁽²⁾	I/O, Fm+ capable
	_u ⁽³⁾	I/O, with USB function
Notes	Unless otherwise specified by a note, all I/Os are set as floating inputs during and after reset	
Pin functions	Alternate functions	Functions selected through GPIOx_AFR registers
	Additional functions	Functions directly selected/enabled through peripheral registers

1. The related I/O structures in [Table 12](#) are: FT_a, FT_fa, TT_a.

2. The related I/O structures in [Table 12](#) are: FT_f, FT_fa.

3. The related I/O structures in [Table 12](#) are FT_u.

Table 12. STM32G431x6/x8/xB pin definition⁽¹⁾

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
-	-	-	-	-	-	-	-	1	PE2	I/O	FT	-	TRACECK, TIM3_CH1, SAI1_CK1, SAI1_MCLK_A, EVENTOUT	-
-	-	-	-	-	-	-	-	2	PE3	I/O	FT	-	TRACED0, TIM3_CH2, SAI1_SD_B, EVENTOUT	-
-	-	-	-	-	-	-	-	3	PE4	I/O	FT	-	TRACED1, TIM3_CH3, SAI1_D2, SAI1_FS_A, EVENTOUT	-
-	-	-	-	-	-	-	-	4	PE5	I/O	FT	-	TRACED2, TIM3_CH4, SAI1_CK2, SAI1_SCK_A, EVENTOUT	-
-	-	-	-	-	-	-	-	5	PE6	I/O	FT	-	TRACED3, SAI1_D1, SAI1_SD_A, EVENTOUT	WKUP3, RTC_TAMP3
-	-	1	1	B7	1	C2	1	6	VBAT	S	-	-	-	-
-	-	2	2	C5	2	B1	2	7	PC13	I/O	FT	(2) (3)	TIM1_BKIN, TIM1_CH1N, TIM8_CH4N, EVENTOUT	WKUP2, RTC_TAMP1, RTC_TS, RTC_OUT1
-	-	3	3	C7	3	C1	3	8	PC14- OSC32_IN	I/O	FT	(2) (3)	EVENTOUT	OSC32_IN
-	-	4	4	D7	4	D1	4	9	PC15- OSC32_OUT	I/O	FT	(2) (3)	EVENTOUT	OSC32_OUT
-	-	-	-	-	-	-	-	10	PF9	I/O	FT	-	TIM15_CH1, SPI2_SCK, SAI1_FS_B, EVENTOUT	-
-	-	-	-	-	-	-	-	11	PF10	I/O	FT	-	TIM15_CH2, SPI2_SCK, SAI1_D3, EVENTOUT	-
2	2	5	5	E7	5	E1	5	12	PF0-OSC_IN	I	FT_fa	-	I2C2_SDA, SPI2_NSS/ I2S2_WS, TIM1_CH3N, EVENTOUT	ADC1_IN10, OSC_IN
3	3	6	6	E6	6	F1	6	13	PF1- OSC_OUT	O	FT_a	-	SPI2_SCK/ I2S2_CK, EVENTOUT	ADC2_IN10, COMP3_INM, OSC_OUT
4	4	7	7	C6	7	D2	7	14	PG10-NRST	I/O	FT	-	MCO, EVENTOUT	NRST

Table 12. STM32G431x6/x8/xB pin definition⁽¹⁾ (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
-	-	-	-	-	8	E2	8	15	PC0	I/O	FT_a	-	LPTIM1_IN1, TIM1_CH1, LPUART1_RX, EVENTOUT	ADC12_IN6, COMP3_INM
-	-	-	-	-	9	C3	9	16	PC1	I/O	TT_a	-	LPTIM1_OUT, TIM1_CH2, LPUART1_TX, SAI1_SD_A, EVENTOUT	ADC12_IN7, COMP3_INP
-	-	-	-	-	10	D3	10	17	PC2	I/O	FT_a	-	LPTIM1_IN2, TIM1_CH3, COMP3_OUT, EVENTOUT	ADC12_IN8
-	-	-	-	-	11	G1	11	18	PC3	I/O	FT_a	-	LPTIM1_ETR, TIM1_CH4, SAI1_D1, TIM1_BKIN2, SAI1_SD_A, EVENTOUT	ADC12_IN9
-	-	-	-	-	-	-	-	19	PF2	I/O	FT	-	I2C2_SMBA, EVENTOUT	-
5	5	8	8	F7	12	F2	12	20	PA0	I/O	TT_a	-	TIM2_CH1, USART2_CTS, COMP1_OUT, TIM8_BKIN, TIM8_ETR, TIM2_ETR, EVENTOUT	ADC12_IN1, COMP1_INM, COMP3_INP, RTC_TAMP2, WKUP1
6	6	9	9	D6	13	E3	13	21	PA1	I/O	TT_a	-	RTC_REFIN, TIM2_CH2, USART2_RTS_DE, TIM15_CH1N, EVENTOUT	ADC12_IN2, COMP1_INP, OPAMP1_VINP, OPAMP3_VINP
7	7	10	10	F6	14	F3	14	22	PA2	I/O	TT_a	-	TIM2_CH3, USART2_TX, COMP2_OUT, TIM15_CH1, LPUART1_TX, UCPD1_FRSTX, EVENTOUT	ADC1_IN3, COMP2_INM, OPAMP1_VOUT, WKUP4/ LSCO
-	-	-	-	-	15	G2	15	23	VSS_2	S	-	-	-	-
-	-	-	-	-	16	H1	16	24	VDD_2	S	-	-	-	-
8	8	11	11	G7	17	H2	17	25	PA3	I/O	TT_a	-	TIM2_CH4, SAI1_CK1, USART2_RX, TIM15_CH2, LPUART1_RX, SAI1_MCLK_A, EVENTOUT	ADC1_IN4, COMP2_INP, OPAMP1_VINM/ OPAMP1_VINP

Table 12. STM32G431x6/x8/xB pin definition⁽¹⁾ (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
9	9	12	12	E5	18	D4	18	26	PA4	I/O	TT_a	-	TIM3_CH2, SPI1_NSS, SPI3_NSS/ I2S3_WS, USART2_CK, SAI1_FS_B, EVENTOUT	ADC2_IN17, DAC1_OUT1, COMP1_INM
10	10	13	13	F5	19	E4	19	27	PA5	I/O	TT_a	-	TIM2_CH1, TIM2_ETR, SPI1_SCK, UCPD1_FRSTX, EVENTOUT	ADC2_IN13, DAC1_OUT2, COMP2_INM, OPAMP2_VINM
11	11	14	14	G6	20	G3	20	28	PA6	I/O	TT_a	-	TIM16_CH1, TIM3_CH1, TIM8_BKIN, SPI1_MISO, TIM1_BKIN, COMP1_OUT, LPUART1_CTS, EVENTOUT	ADC2_IN3, OPAMP2_VOUT
12	12	15	15	D5	21	H3	21	29	PA7	I/O	TT_a	-	TIM17_CH1, TIM3_CH2, TIM8_CH1N, SPI1_MOSI, TIM1_CH1N, COMP2_OUT, UCPD1_FRSTX, EVENTOUT	ADC2_IN4, COMP2_INP, OPAMP1_VINP, OPAMP2_VINP
-	-	16	-	D4	22	D5	22	30	PC4	I/O	FT_fa	-	TIM1_ETR, I2C2_SCL, USART1_TX, EVENTOUT	ADC2_IN5
-	-	-	-	-	23	F4	23	31	PC5	I/O	TT_a	-	TIM15_BKIN, SAI1_D3, TIM1_CH4N, USART1_RX, EVENTOUT	ADC2_IN11, OPAMP1_VINM, OPAMP2_VINM, WKUP5
13	13	17	16	G5	24	E5	24	32	PB0	I/O	TT_a	-	TIM3_CH3, TIM8_CH2N, TIM1_CH2N, UCPD1_FRSTX, EVENTOUT	ADC1_IN15, COMP4_INP, OPAMP2_VINP, OPAMP3_VINP
-	-	18	17	E4	25	F5	25	33	PB1	I/O	TT_a	-	TIM3_CH4, TIM8_CH3N, TIM1_CH3N, COMP4_OUT, LPUART1_RTS_DE, EVENTOUT	ADC1_IN12, COMP1_INP, OPAMP3_VOUT
-	-	19	18	F4	26	H4	26	34	PB2	I/O	TT_a	-	RTC_OUT2, LPTIM1_OUT, I2C3_SMB_A, EVENTOUT	ADC2_IN12, COMP4_INM, OPAMP3_VINM

Table 12. STM32G431x6/x8/xB pin definition⁽¹⁾ (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
14	14	-	19	G4	27	G4	27	35	VSSA	S	-	-	-	-
-	-	20	20	F3	28	G5	28	36	VREF+	S	-	-	-	VREFBUF_OUT
15	15	21	21	G3	29	H5	29	37	VDDA	S	-	-	-	-
-	-	-	-	-	-	-	-	30	38	PE7	I/O	TT_a	TIM1_ETR, SAI1_SD_B, EVENTOUT	COMP4_INP
-	-	-	-	-	-	-	-	31	39	PE8	I/O	FT_a	TIM1_CH1N, SAI1_SCK_B, EVENTOUT	COMP4_INM
-	-	-	-	-	-	-	-	32	40	PE9	I/O	FT	TIM1_CH1, SAI1_FS_B, EVENTOUT	-
-	-	-	-	-	-	-	-	33	41	PE10	I/O	FT	TIM1_CH2N, SAI1_MCLK_B, EVENTOUT	-
-	-	-	-	-	-	-	-	34	42	PE11	I/O	FT	TIM1_CH2, EVENTOUT	-
-	-	-	-	-	-	-	-	35	43	PE12	I/O	FT	TIM1_CH3N, EVENTOUT	-
-	-	-	-	-	-	-	-	36	44	PE13	I/O	FT	TIM1_CH3, EVENTOUT	-
-	-	-	-	-	-	-	-	37	45	PE14	I/O	FT	TIM1_CH4, TIM1_BKIN2, EVENTOUT	-
-	-	-	-	-	-	-	-	38	46	PE15	I/O	FT	TIM1_BKIN, TIM1_CH4N, USART3_RX, EVENTOUT	-
-	-	22	22	F2	30	H6	39	47	PB10	I/O	TT_a	-	TIM2_CH3, USART3_TX, LPUART1_RX, TIM1_BKIN, SAI1_SCK_A, EVENTOUT	OPAMP3_VINM
16	16	-	23	G2	31	G7	40	48	VSS	S	-	-	-	-
17	17	23	24	G1	32	H8	41	49	VDD	S	-	-	-	-
-	-	24	25	E3	33	H7	42	50	PB11	I/O	FT_a	-	TIM2_CH4, USART3_RX, LPUART1_TX, EVENTOUT	ADC12_IN14

Table 12. STM32G431x6/x8/xB pin definition⁽¹⁾ (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
-	-	25	26	F1	34	G8	43	51	PB12	I/O	FT_a	-	I2C2_SMBA, SPI2_NSS/ I2S2_WS, TIM1_BKIN, USART3_CK, LPUART1_RTS_DE, EVENTOUT	ADC1_IN11
-	-	26	27	E2	35	G6	44	52	PB13	I/O	TT_a	-	SPI2_SCK/I2S2_CK, TIM1_CH1N, USART3_CTS, LPUART1_CTS, EVENTOUT	OPAMP3_VINP
-	-	27	28	D3	36	F8	45	53	PB14	I/O	TT_a	-	TIM15_CH1, SPI2_MISO, TIM1_CH2N, USART3_RTS_DE, COMP4_OUT, EVENTOUT	ADC1_IN5, OPAMP2_VINP
-	-	28	29	E1	37	F7	46	54	PB15	I/O	FT_a	-	RTC_REFIN, TIM15_CH2, TIM15_CH1N, COMP3_OUT, TIM1_CH3N, SPI2_MOSI/ I2S2_SD, EVENTOUT	ADC2_IN15
-	-	-	-	-	-	-	47	55	PD8	I/O	FT_a	-	USART3_TX, EVENTOUT	-
-	-	-	-	-	-	-	48	56	PD9	I/O	FT	-	USART3_RX, EVENTOUT	-
-	-	-	-	-	-	-	49	57	PD10	I/O	FT	-	USART3_CK, EVENTOUT	-
-	-	-	-	-	-	-	-	58	PD11	I/O	FT_a	-	USART3_CTS, EVENTOUT	-
-	-	-	-	-	-	-	-	59	PD12	I/O	TT	-	TIM4_CH1, USART3_RTS_DE, EVENTOUT	-
-	-	-	-	-	-	-	-	60	PD13	I/O	FT	-	TIM4_CH2, EVENTOUT	-
-	-	-	-	-	-	-	-	61	PD14	I/O	FT_a	-	TIM4_CH3, EVENTOUT	OPAMP2_VINP
-	-	-	-	-	-	-	-	62	PD15	I/O	FT	-	TIM4_CH4, SPI2_NSS, EVENTOUT	-
-	-	-	-	-	-	-	50	63	VSS	S	-	-	-	-
-	-	-	-	-	-	-	51	64	VDD	S	-	-	-	-

Table 12. STM32G431x6/x8/xB pin definition⁽¹⁾ (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
-	-	29	-	D1	38	E8	52	65	PC6	I/O	FT	-	TIM3_CH1, TIM8_CH1, I2S2_MCK, EVENTOUT	-
-	-	-	-	-	39	E7	53	66	PC7	I/O	FT	-	TIM3_CH2, TIM8_CH2, I2S3_MCK, EVENTOUT	-
-	-	-	-	-	40	F6	54	67	PC8	I/O	FT_f	-	TIM3_CH3, TIM8_CH3, I2C3_SCL, EVENTOUT	-
-	-	-	-	-	41	D8	55	68	PC9	I/O	FT_f	-	TIM3_CH4, TIM8_CH4, I2SCKIN, TIM8_BKIN2, I2C3_SDA, EVENTOUT	-
18	18	30	30	D2	42	E6	56	69	PA8	I/O	FT_f	-	MCO, I2C3_SCL, I2C2_SDA, I2S2_MCK, TIM1_CH1, USART1_CK, TIM4_ETR, SAI1_CK2, SAI1_SCK_A, EVENTOUT	-
19	19	31	31	C3	43	D7	57	70	PA9	I/O	FT_fd	-	I2C3_SMBA, I2C2_SCL, I2S3_MCK, TIM1_CH2, USART1_TX, TIM15_BKIN, TIM2_CH3, SAI1_FS_A, EVENTOUT	UCPD1_DBCC1
20	20	32	32	C2	44	D6	58	71	PA10	I/O	FT_d a	-	TIM17_BKIN, USB_CRD_SYNC, I2C2_SMBA, SPI2_MISO, TIM1_CH3, USART1_RX, TIM2_CH4, TIM8_BKIN, SAI1_D1, SAI1_SD_A, EVENTOUT	UCPD1_DBCC2
21	21	33	33	C1	45	C8	59	72	PA11	I/O	FT_u	-	SPI2_MOSI/ I2S2_SD, TIM1_CH1N, USART1_CTS, COMP1_OUT, FDCAN1_RX, TIM4_CH1, TIM1_CH4, TIM1_BKIN2, EVENTOUT	USB_DM

Table 12. STM32G431x6/x8/xB pin definition⁽¹⁾ (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
22	22	34	34	B1	46	B8	60	73	PA12	I/O	FT_u	-	TIM16_CH1, I2SCKIN, TIM1_CH2N, USART1_RTS_DE, COMP2_OUT, FDCAN1_TX, TIM4_CH2, TIM1_ETR, EVENTOUT	USB_DP
-	-	-	35	-	47	B7	61	74	VSS	S	-	-	-	-
-	-	35	36	-	48	A8	62	75	VDD	S	-	-	-	-
23	23	36	37	B2	49	C7	63	76	PA13	I/O	FT_f	(4)	SWDIO-JTMS, TIM16_CH1N, I2C1_SCL, IR_OUT, USART3_CTS, TIM4_CH3, SAI1_SD_B, EVENTOUT	-
24	24	37	38	B3	50	C6	64	77	PA14	I/O	FT_f	(4)	SWCLK-JTCK, LPTIM1_OUT, I2C1_SDA, TIM8_CH2, TIM1_BKIN, USART2_TX, SAI1_FS_B, EVENTOUT	-
25	25	38	39	A1	51	A7	65	78	PA15	I/O	FT_f	(4)	JTDI, TIM2_CH1, TIM8_CH1, I2C1_SCL, SPI1_NSS, SPI3_NSS/ I2S3_WS, USART2_RX, UART4_RTS_DE, TIM1_BKIN, TIM2_ETR, EVENTOUT	-
-	-	39	-	-	52	C5	66	79	PC10	I/O	FT	-	TIM8_CH1N, UART4_TX, SPI3_SCK/ I2S3_CK, USART3_TX, EVENTOUT	-
-	-	40	-	A2	53	B6	67	80	PC11	I/O	FT_f	-	TIM8_CH2N, UART4_RX, SPI3_MISO, USART3_RX, I2C3_SDA, EVENTOUT	-

Table 12. STM32G431x6/x8/xB pin definition⁽¹⁾ (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
-	-	-	-	-	54	A6	68	81	PC12	I/O	FT	-	TIM8_CH3N, SPI3_MOSI/ I2S3_SD, USART3_CK, UCPD1_FRSTX, EVENTOUT	-
-	-	-	-	-	-	-	69	82	PD0	I/O	FT	-	TIM8_CH4N, FDCAN1_RX, EVENTOUT	-
-	-	-	-	-	-	-	70	83	PD1	I/O	FT	-	TIM8_CH4, TIM8_BKIN2, FDCAN1_TX, EVENTOUT	-
-	-	-	-	-	55	B5	71	84	PD2	I/O	FT	-	TIM3_ETR, TIM8_BKIN, EVENTOUT	-
-	-	-	-	-	-	-	-	85	PD3	I/O	FT	-	TIM2_CH1/ TIM2_ETR, USART2_CTS, EVENTOUT	-
-	-	-	-	-	-	-	-	86	PD4	I/O	FT	-	TIM2_CH2, USART2_RTS_DE, EVENTOUT	-
-	-	-	-	-	-	-	-	87	PD5	I/O	FT	-	USART2_TX, EVENTOUT	-
-	-	-	-	-	-	-	-	88	PD6	I/O	FT	-	TIM2_CH4, SAI1_D1, USART2_RX, SAI1_SD_A, EVENTOUT	-
-	-	-	-	-	-	-	-	89	PD7	I/O	FT	-	TIM2_CH3, USART2_CK, EVENTOUT	-
26	26	41	40	A3	56	A5	72	90	PB3	I/O	FT	(4)	JTDO-TRACESWO, TIM2_CH2, TIM4_ETR, USB_CRD_SYNC, TIM8_CH1N, SPI1_SCK, SPI3_SCK/ I2S3_CK, USART2_TX, TIM3_ETR, SAI1_SCK_B, EVENTOUT	-

Table 12. STM32G431x6/x8/xB pin definition⁽¹⁾ (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSP49	LQFP64	UFBGA64	LQFP80	LQFP100						
27	27	42	41	B4	57	C4	73	91	PB4	I/O	FT_c	(4)	JTRST, TIM16_CH1, TIM3_CH1, TIM8_CH2N, SPI1_MISO, SPI3_MISO, USART2_RX, TIM17_BKIN, SAI1_MCLK_B, EVENTOUT	UCPD1_CC2
28	28	43	42	A43	58	B4	74	92	PB5	I/O	FT_f	-	TIM16_BKIN, TIM3_CH2, TIM8_CH3N, I2C1_SMB_A, SPI1_MOSI, SPI3_MOSI/ I2S3_SD, USART2_CK, I2C3_SDA, TIM17_CH1, LPTIM1_IN1, SAI1_SD_B, EVENTOUT	-
29	29	44	43	C4	59	A4	75	93	PB6	I/O	FT_c	-	TIM16_CH1N, TIM4_CH1, TIM8_CH1, TIM8_ETR, USART1_TX, COMP4_OUT, TIM8_BKIN2, LPTIM1_ETR, SAI1_FS_B, EVENTOUT	UCPD1_CC1
30	30	45	44	A5	60	A3	76	94	PB7	I/O	FT_f	-	TIM17_CH1N, TIM4_CH2, I2C1_SDA, TIM8_BKIN, USART1_RX, COMP3_OUT, TIM3_CH4, LPTIM1_IN2, UART4_CTS, EVENTOUT	PVD_IN
31	31	46	45	B5	61	B3	77	95	PB8-BOOT0	I/O	FT_f	(5)	TIM16_CH1, TIM4_CH3, SAI1_CK1, I2C1_SCL, USART3_RX, COMP1_OUT, FDCAN1_RX, TIM8_CH2, TIM1_BKIN, SAI1_MCLK_A, EVENTOUT	-

Table 12. STM32G431x6/x8/xB pin definition⁽¹⁾ (continued)

Pin Number									Pin name (function after reset)	Pin type	I/O structure	Notes	Alternate function	Additional functions
UFQFPN32	LQFP32	UFQFPN48	LQFP48	WLCSFP49	LQFP64	UFBGA64	LQFP80	LQFP100						
-	-	47	46	B6	62	A2	78	96	PB9	I/O	FT_f	-	TIM17_CH1, TIM4_CH4, SAI1_D2, I2C1_SDA, IR_OUT, USART3_TX, COMP2_OUT, FDCAN1_TX, TIM8_CH3, TIM1_CH3N, SAI1_FS_A, EVENTOUT	-
-	-	-	-	-	-	-	-	97	PE0	I/O	FT	-	TIM4_ETR, TIM16_CH1, USART1_TX, EVENTOUT	-
-	-	-	-	-	-	-	-	98	PE1	I/O	FT	-	TIM17_CH1, USART1_RX, EVENTOUT	-
32	32	-	47	A6	63	B2	79	99	VSS	S	-	-	-	-
1	1	48	48	A7	64	A1	80	100	VDD	S	-	-	-	-

- Function availability depends on the chosen device.
- PC13, PC14 and PC15 are supplied through the power switch. Since the switch only sinks a limited amount of current (3 mA), the use of GPIOs PC13 to PC15 in output mode is limited:
 - The speed should not exceed 2 MHz with a maximum load of 30 pF
 - These GPIOs must not be used as current sources (e.g. to drive an LED).
- After a Backup domain power-up, PC13, PC14 and PC15 operate as GPIOs. Their function then depends on the content of the RTC registers which are not reset by the system reset. For details on how to manage these GPIOs, refer to the Backup domain and RTC register descriptions in the reference manual RM0440 "STM32G4 Series advanced Arm®-based 32-bit MCUs".
- After reset, these pins are configured as JTAG/SW debug alternate functions, and the internal pull-up on PA15, PA13, PB4 pins and the internal pull-down on PA14 pin are activated.
- It is recommended to set PB8 in another mode than analog mode after startup to limit consumption if the pin is left unconnected.

4.11 Alternate functions

Table 13. Alternate function

Port		AF0	AF1	AF2	AF3	AF4	AF5	AF6	AF7	AF8	AF9	AF10	AF11	AF12	AF13	AF14	AF15
		SYS_AF	LPTIM1/TIM2/5/15/16/17	I2C3/TIM1/2/3/4/8/15/GPCOMP1	I2C3/SAI1/USB/TIM8/15/GPCOMP3	I2C1/2/3/TIM1/8/16/17	SPI1/2/3/I2S2/3/UART4/TIM8/Infra red	SPI2/3/I2S2/3/TIM1/8/Infrared	USART1/2/3	I2C3/4/UART4/LPUART1/GPCOMP1/2/3	TIM1/8/15/FDCAN1	TIM2/3/4/8/17	LPTIM1/TIM1/8/FDCAN1	LPUART1/SAI1/TIM1	SAI1/OPAMP2	UART4/SAI1/TIM2/15/UCPD1	EVENT
Port A	PA0	-	TIM2_CH1	-	-	-	-	-	USART2_CTS	COMP1_OUT	TIM8_BKIN	TIM8_ETR	-	-	-	TIM2_ETR	EVENT OUT
	PA1	RTC_REFIN	TIM2_CH2	-	-	-	-	-	USART2_RTS_DE	-	TIM15_CH1N	-	-	-	-	-	EVENT OUT
	PA2	-	TIM2_CH3	-	-	-	-	-	USART2_TX	COMP2_OUT	TIM15_CH1	-	-	LPUART1_TX	-	UCPD1_FRSTX	EVENT OUT
	PA3	-	TIM2_CH4	-	SAI1_CK1	-	-	-	USART2_RX	-	TIM15_CH2	-	-	LPUART1_RX	SAI1_MCLK_A	-	EVENT OUT
	PA4	-	-	TIM3_CH2	-	-	SPI1_NSS	SPI3_NSS/I2S3_WS	USART2_CK	-	-	-	-	-	SAI1_FS_B	-	EVENT OUT
	PA5	-	TIM2_CH1	TIM2_ETR	-	-	SPI1_SCK	-	-	-	-	-	-	-	-	UCPD1_FRSTX	EVENT OUT
	PA6	-	TIM16_CH1	TIM3_CH1	-	TIM8_BKIN	SPI1_MISO	TIM1_BKIN	-	COMP1_OUT	-	-	-	LPUART1_CTS	-	-	EVENT OUT
	PA7	-	TIM17_CH1	TIM3_CH2	-	TIM8_CH1N	SPI1_MOSI	TIM1_CH1N	-	COMP2_OUT	-	-	-	-	-	UCPD1_FRSTX	EVENT OUT
	PA8	MCO	-	I2C3_SCL	-	I2C2_SDA	I2S2_MCK	TIM1_CH1	USART1_CK	-	-	TIM4_ETR	-	SAI1_CK2	-	SAI1_SCK_A	EVENT OUT
	PA9	-	-	I2C3_SMBA	-	I2C2_SCL	I2S3_MCK	TIM1_CH2	USART1_TX	-	TIM15_BKIN	TIM2_CH3	-	-	-	SAI1_FS_A	EVENT OUT
	PA10	-	TIM17_BKIN	-	USB_CR2_SYNC	I2C2_SMBA	SPI2_MISO	TIM1_CH3	USART1_RX	-	-	TIM2_CH4	TIM8_BKIN	SAI1_D1	-	SAI1_SD_A	EVENT OUT
	PA11	-	-	-	-	-	SPI2_MOSI/I2S2_SD	TIM1_CH1N	USART1_CTS	COMP1_OUT	FDCAN1_RX	TIM4_CH1	TIM1_CH4	TIM1_BKIN2	-	-	EVENT OUT
	PA12	-	TIM16_CH1	-	-	-	I2SCKIN	TIM1_CH2N	USART1_RTS_DE	COMP2_OUT	FDCAN1_TX	TIM4_CH2	TIM1_ETR	-	-	-	EVENT OUT
	PA13	SWDIO-JTMS	TIM16_CH1N	-	-	I2C1_SCL	IR_OUT	-	USART3_CTS	-	-	TIM4_CH3	-	-	SAI1_SD_B	-	EVENT OUT
	PA14	SWCLK-JTCK	LPTIM1_OUT	-	-	I2C1_SDA	TIM8_CH2	TIM1_BKIN	USART2_TX	-	-	-	-	-	SAI1_FS_B	-	EVENT OUT
	PA15	JTDI	TIM2_CH1	TIM8_CH1	-	I2C1_SCL	SPI1_NSS	SPI3_NSS/I2S3_WS	USART2_RX	UART4_RTS_DE	TIM1_BKIN	-	-	-	-	TIM2_ETR	EVENT OUT



Table 13. Alternate function (continued)

Port		AF0	AF1	AF2	AF3	AF4	AF5	AF6	AF7	AF8	AF9	AF10	AF11	AF12	AF13	AF14	AF15
		SYS_AF	LPTIM1/TIM2/5/15/16/17	I2C3/TIM1/2/3/4/8/15/GPCOMP1	I2C3/SAI1/USB/TIM8/15/GPCOMP3	I2C1/2/3/TIM1/8/16/17	SPI1/2/3/I2S2/3/UART4/TIM8/Infra red	SPI2/3/I2S2/3/TIM1/8/Infrared	USART1/2/3	I2C3/4/UART4/LPUART1/GPCOMP1/2/3	TIM1/8/15/FDCAN1	TIM2/3/4/8/17	LPTIM1/TIM1/8/FDCAN1	LPUART1/SAI1/TIM1	SAI1/OPAMP2	UART4/SAI1/TIM2/15/UCPD1	EVENT
Port B	PB0	-	-	TIM3_CH3	-	TIM8_CH2N	-	TIM1_CH2N	-	-	-	-	-	-	-	UCPD1_FRSTX	EVENT OUT
	PB1	-	-	TIM3_CH4	-	TIM8_CH3N	-	TIM1_CH3N	-	COMP4_OUT	-	-	-	LPUART1_RTS_DE	-	-	EVENT OUT
	PB2	RTC_OUT2	LPTIM1_OUT	-	-	I2C3_SMBA	-	-	-	-	-	-	-	-	-	-	EVENT OUT
	PB3	JTDO-TRACESWO	TIM2_CH2	TIM4_ETR	USB_CR_S_SYNC	TIM8_CH1N	SPI1_SCK	SPI3_SCK/I2S3_CK	USART2_TX	-	-	TIM3_ETR	-	-	-	SAI1_SCK_B	EVENT OUT
	PB4	JTRST	TIM16_CH1	TIM3_CH1	-	TIM8_CH2N	SPI1_MISO	SPI3_MISO	USART2_RX	-	-	TIM17_BKIN	-	-	-	SAI1_MCLK_B	EVENT OUT
	PB5	-	TIM16_BKIN	TIM3_CH2	TIM8_CH3N	I2C1_SMBA	SPI1_MOSI	SPI3_MOSI/I2S3_SD	USART2_CK	I2C3_SDA	-	TIM17_CH1	LPTIM1_IN1	SAI1_SD_B	-	-	EVENT OUT
	PB6	-	TIM16_CH1N	TIM4_CH1	-	-	TIM8_CH1	TIM8_ETR	USART1_TX	COMP4_OUT	-	TIM8_BKIN2	LPTIM1_ETR	-	-	SAI1_FS_B	EVENT OUT
	PB7	-	TIM17_CH1N	TIM4_CH2	-	I2C1_SDA	TIM8_BKIN	-	USART1_RX	COMP3_OUT	-	TIM3_CH4	LPTIM1_IN2	-	-	UART4_CTS	EVENT OUT
	PB8	-	TIM16_CH1	TIM4_CH3	SAI1_CK1	I2C1_SCL	-	-	USART3_RX	COMP1_OUT	FDCAN1_RX	TIM8_CH2	-	TIM1_BKIN	-	SAI1_MCLK_A	EVENT OUT
	PB9	-	TIM17_CH1	TIM4_CH4	SAI1_D2	I2C1_SDA	-	IR_OUT	USART3_TX	COMP2_OUT	FDCAN1_TX	TIM8_CH3	-	TIM1_CH3N	-	SAI1_FS_A	EVENT OUT
	PB10	-	TIM2_CH3	-	-	-	-	-	USART3_TX	LPUART1_RX	-	-	-	TIM1_BKIN	-	SAI1_SCK_A	EVENT OUT
	PB11	-	TIM2_CH4	-	-	-	-	-	USART3_RX	LPUART1_TX	-	-	-	-	-	-	EVENT OUT
	PB12	-	-	-	-	I2C2_SMBA	SPI2_NSS/I2S2_WS	TIM1_BKIN	USART3_CK	LPUART1_RTS_DE	-	-	-	-	-	-	EVENT OUT
	PB13	-	-	-	-	-	SPI2_SCK/I2S2_CK	TIM1_CH1N	USART3_CTS	LPUART1_CTS	-	-	-	-	-	-	EVENT OUT
	PB14	-	TIM15_CH1	-	-	-	SPI2_MISO	TIM1_CH2N	USART3_RTS_DE	COMP4_OUT	-	-	-	-	-	-	EVENT OUT
	PB15	RTC_REFIN	TIM15_CH2	TIM15_CH1N	COMP3_OUT	TIM1_CH3N	SPI2_MOSI/I2S2_SD	-	-	-	-	-	-	-	-	-	EVENT OUT

Table 13. Alternate function (continued)

Port		AF0	AF1	AF2	AF3	AF4	AF5	AF6	AF7	AF8	AF9	AF10	AF11	AF12	AF13	AF14	AF15
		SYS_AF	LPTIM1/TIM2/5/15/16/17	I2C3/TIM1/2/3/4/8/15/GPCOMP1	I2C3/SAI1/USB/TIM8/15/GPCOMP3	I2C1/2/3/TIM1/8/16/17	SPI1/2/3/I2S2/3/UART4/TIM8/Infra red	SPI2/3/I2S2/3/TIM1/8/Infra red	USART1/2/3	I2C3/4/UART4/LPUART1/GPCOMP1/2/3	TIM1/8/15/FDCAN1	TIM2/3/4/8/17	LPTIM1/TIM1/8/FDCA N1	LPUART1/SAI1/TIM1	SAI1/OPAMP2	UART4/SAI1/TIM2/15/UCPD1	EVENT
Port C	PC0	-	LPTIM1_IN1	TIM1_CH1	-	-	-	-	-	LPUART1_RX	-	-	-	-	-	-	EVENT OUT
	PC1	-	LPTIM1_OUT	TIM1_CH2	-	-	-	-	-	LPUART1_TX	-	-	-	-	SAI1_SD_A	-	EVENT OUT
	PC2	-	LPTIM1_IN2	TIM1_CH3	COMP3_OUT	-	-	-	-	-	-	-	-	-	-	-	EVENT OUT
	PC3	-	LPTIM1_ETR	TIM1_CH4	SAI1_D1	-	-	TIM1_BKIN2	-	-	-	-	-	-	SAI1_SD_A	-	EVENT OUT
	PC4	-	-	TIM1_ETR	-	I2C2_SCL	-	-	USART1_TX	-	-	-	-	-	-	-	EVENT OUT
	PC5	-	-	TIM15_BKIN	SAI1_D3	-	-	TIM1_CH4N	USART1_RX	-	-	-	-	-	-	-	EVENT OUT
	PC6	-	-	TIM3_CH1	-	TIM8_CH1	-	I2S2_MCK	-	-	-	-	-	-	-	-	EVENT OUT
	PC7	-	-	TIM3_CH2	-	TIM8_CH2	-	I2S3_MCK	-	-	-	-	-	-	-	-	EVENT OUT
	PC8	-	-	TIM3_CH3	-	TIM8_CH3	-	-	-	I2C3_SCL	-	-	-	-	-	-	EVENT OUT
	PC9	-	-	TIM3_CH4	-	TIM8_CH4	I2SCKIN	TIM8_BKIN2	-	I2C3_SDA	-	-	-	-	-	-	EVENT OUT
	PC10	-	-	-	-	TIM8_CH1N	UART4_TX	SPI3_SCK/I2S3_CK	USART3_TX	-	-	-	-	-	-	-	EVENT OUT
	PC11	-	-	-	-	TIM8_CH2N	UART4_RX	SPI3_MISO	USART3_RX	I2C3_SDA	-	-	-	-	-	-	EVENT OUT
	PC12	-	-	-	-	TIM8_CH3N	-	SPI3_MOSI/I2S3_SD	USART3_CK	-	-	-	-	-	-	UCPD1_FRSTX	EVENT OUT
	PC13	-	-	TIM1_BKIN	-	TIM1_CH1N	-	TIM8_CH4N	-	-	-	-	-	-	-	-	EVENT OUT
	PC14	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	EVENT OUT
	PC15	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	EVENT OUT



Table 13. Alternate function (continued)

Port		AF0	AF1	AF2	AF3	AF4	AF5	AF6	AF7	AF8	AF9	AF10	AF11	AF12	AF13	AF14	AF15
		SYS_AF	LPTIM1/TIM2/5/15/16/17	I2C3/TIM1/2/3/4/8/15/GPCOMP1	I2C3/SAI1/USB/TIM8/15/GPCOMP3	I2C1/2/3/TIM1/8/16/17	SPI1/2/3/I2S2/3/UART4/TIM8/Infra red	SPI2/3/I2S2/3/TIM1/8/Infra red	USART1/2/3	I2C3/4/UART4/LPUART1/GPCOMP1/2/3	TIM1/8/15/FDCAN1	TIM2/3/4/8/17	LPTIM1/TIM1/8/FDCA N1	LPUART1/SAI1/TIM1	SAI1/OPAMP2	UART4/SAI1/TIM2/15/UCPD1	EVENT
Port D	PD0	-	-	-	-	-	-	TIM8_CH4N	-	-	FDCAN1_RX	-	-	-	-	-	EVENT OUT
	PD1	-	-	-	-	TIM8_CH4	-	TIM8_BKIN2	-	-	FDCAN1_TX	-	-	-	-	-	EVENT OUT
	PD2	-	-	TIM3_ETR	-	TIM8_BKIN	-	-	-	-	-	-	-	-	-	-	EVENT OUT
	PD3	-	-	TIM2_CH1/TIM2_ETR	-	-	-	-	USART2_CTS	-	-	-	-	-	-	-	EVENT OUT
	PD4	-	-	TIM2_CH2	-	-	-	-	USART2_RTS_DE	-	-	-	-	-	-	-	EVENT OUT
	PD5	-	-	-	-	-	-	-	USART2_TX	-	-	-	-	-	-	-	EVENT OUT
	PD6	-	-	TIM2_CH4	SAI1_D1	-	-	-	USART2_RX	-	-	-	-	-	SAI1_SD_A	-	EVENT OUT
	PD7	-	-	TIM2_CH3	-	-	-	-	USART2_CK	-	-	-	-	-	-	-	EVENT OUT
	PD8	-	-	-	-	-	-	-	USART3_TX	-	-	-	-	-	-	-	EVENT OUT
	PD9	-	-	-	-	-	-	-	USART3_RX	-	-	-	-	-	-	-	EVENT OUT
	PD10	-	-	-	-	-	-	-	USART3_CK	-	-	-	-	-	-	-	EVENT OUT
	PD11	-	-	-	-	-	-	-	USART3_CTS	-	-	-	-	-	-	-	EVENT OUT
	PD12	-	-	TIM4_CH1	-	-	-	-	USART3_RTS_DE	-	-	-	-	-	-	-	EVENT OUT
	PD13	-	-	TIM4_CH2	-	-	-	-	-	-	-	-	-	-	-	-	EVENT OUT
	PD14	-	-	TIM4_CH3	-	-	-	-	-	-	-	-	-	-	-	-	EVENT OUT
	PD15	-	-	TIM4_CH4	-	-	-	SPI2_NSS	-	-	-	-	-	-	-	-	EVENT OUT

Table 13. Alternate function (continued)

Port		AF0	AF1	AF2	AF3	AF4	AF5	AF6	AF7	AF8	AF9	AF10	AF11	AF12	AF13	AF14	AF15
		SYS_AF	LPTIM1/TIM2/5/15/16/17	I2C3/TIM1/2/3/4/8/15/GPCOMP1	I2C3/SAI1/USB/TIM8/15/GPCOMP3	I2C1/2/3/TIM1/8/16/17	SPI1/2/3/I2S2/3/UART4/TIM8/Infrared	SPI2/3/I2S2/3/TIM1/8/Infrared	USART1/2/3	I2C3/4/UART4/LPUART1/GPCOMP1/2/3	TIM1/8/15/FDCAN1	TIM2/3/4/8/17	LPTIM1/TIM1/8/FDCA N1	LPUART1/SAI1/TIM1	SAI1/OPAMP2	UART4/SAI1/TIM2/15/UCPD1	EVENT
Port E	PE0	-	-	TIM4_ETR	-	TIM16_CH1	-	-	USART1_TX	-	-	-	-	-	-	-	EVENT OUT
	PE1	-	-	-	-	TIM17_CH1	-	-	USART1_RX	-	-	-	-	-	-	-	EVENT OUT
	PE2	TRACECK	-	TIM3_CH1	SAI1_CK1	-	-	-	-	-	-	-	-	-	SAI1_MCLK_A	-	EVENT OUT
	PE3	TRACED0	-	TIM3_CH2	-	-	-	-	-	-	-	-	-	-	SAI1_SD_B	-	EVENT OUT
	PE4	TRACED1	-	TIM3_CH3	SAI1_D2	-	-	-	-	-	-	-	-	-	SAI1_FS_A	-	EVENT OUT
	PE5	TRACED2	-	TIM3_CH4	SAI1_CK2	-	-	-	-	-	-	-	-	-	SAI1_SCK_A	-	EVENT OUT
	PE6	TRACED3	-	-	SAI1_D1	-	-	-	-	-	-	-	-	-	SAI1_SD_A	-	EVENT OUT
	PE7	-	-	TIM1_ETR	-	-	-	-	-	-	-	-	-	-	SAI1_SD_B	-	EVENT OUT
	PE8	-	-	TIM1_CH1N	-	-	-	-	-	-	-	-	-	-	SAI1_SCK_B	-	EVENT OUT
	PE9	-	-	TIM1_CH1	-	-	-	-	-	-	-	-	-	-	SAI1_FS_B	-	EVENT OUT
	PE10	-	-	TIM1_CH2N	-	-	-	-	-	-	-	-	-	-	SAI1_MCLK_B	-	EVENT OUT
	PE11	-	-	TIM1_CH2	-	-	-	-	-	-	-	-	-	-	-	-	EVENT OUT
	PE12	-	-	TIM1_CH3N	-	-	-	-	-	-	-	-	-	-	-	-	EVENT OUT
	PE13	-	-	TIM1_CH3	-	-	-	-	-	-	-	-	-	-	-	-	EVENT OUT
	PE14	-	-	TIM1_CH4	-	-	-	TIM1_BKIN2	-	-	-	-	-	-	-	-	EVENT OUT
	PE15	-	-	TIM1_BKIN	-	-	-	TIM1_CH4N	USART3_RX	-	-	-	-	-	-	-	EVENT OUT

Table 13. Alternate function (continued)

Port		AF0	AF1	AF2	AF3	AF4	AF5	AF6	AF7	AF8	AF9	AF10	AF11	AF12	AF13	AF14	AF15
		SYS_AF	LPTIM1/TIM2/5/15/16/17	I2C3/TIM1/2/3/4/8/15/GPCOMP1	I2C3/SAI1/USB/TIM8/15/GPCOMP3	I2C1/2/3/TIM1/8/16/17	SPI1/2/3/I2S2/3/UART4/TIM8/Infrared	SPI2/3/I2S2/3/TIM1/8/Infrared	USART1/2/3	I2C3/4/UART4/LPUART1/GPCOMP1/2/3	TIM1/8/15/FDCAN1	TIM2/3/4/8/17	LPTIM1/TIM1/8/FDCA N1	LPUART1/SAI1/TIM1	SAI1/OPAMP2	UART4/SAI1/TIM2/15/UCPD1	EVENT
Port F	PF0	-	-	-	-	I2C2_SDA	SPI2_NSS/I2S2_WS	TIM1_CH3N	-	-	-	-	-	-	-	-	EVENT OUT
	PF1	-	-	-	-	-	SPI2_SCK/I2S2_CK	-	-	-	-	-	-	-	-	-	EVENT OUT
	PF2	-	-	-	-	I2C2_SMBA	-	-	-	-	-	-	-	-	-	-	EVENT OUT
	PF9	-	-	-	TIM15_CH1	-	SPI2_SCK	-	-	-	-	-	-	-	SAI1_FS_B	-	EVENT OUT
	PF10	-	-	-	TIM15_CH2	-	SPI2_SCK	-	-	-	-	-	-	-	SAI1_D3	-	EVENT OUT
Port G	PG10	MCO	-	-	-	-	-	-	-	-	-	-	-	-	-	-	EVENT OUT